iLearn Practice Exam

10BR · Chapter / Topic 3

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 3: Hardware

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 70

Marks obtained: ..... / 70

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 70.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 13 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, mark your work against the official Cambridge mark scheme (see the source list on the last page).

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 Instructions are processed by a central processing unit (CPU) in a computer.

(a) Complete the paragraph about fetching an instruction into the CPU to be processed.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

address arithmetic logic unit (ALU) binary control unit (CU)

current instruction register (CIR) data denary driver

fetch interrupt memory address register (MAR)

memory data register (MDR) random access memory (RAM)

read only memory (ROM) secondary storage signal

The program counter contains the ……

of the next instruction to be processed; this is then sent to the

…… using the address bus. The address is then

sent to the …… .

Once the address is received, the instruction stored at the location is

sent to the ……, using the

…… bus. The instruction is then

sent to the …… that is built into the

…… .

[7]

(b) The CPU uses an instruction set to decode the instruction.

State what is meant by an instruction set.

.....

..... [1]

2 A computer has both a media access control (MAC) address and an internet protocol (IP) address.

(a) Tick (✓) **one** box to show which of the statements is correct about the MAC address.

A It is assigned by the manufacturer. ☐

B It is assigned by a router. ☐

C It can be static or dynamic. ☐

D It is made up of three different parts. ☐

[1]

(b) An IP address can have an IPv4 or IPv6 format.

(i) Give an example of an IP address that has an IPv4 format.

..... [1]

(ii) Give **two** characteristics of the IPv6 format.

1 .....

.....

2 .....

.....

3 A company uses both solid-state and optical secondary storage.

(a) Explain why a computer needs secondary storage.

.....

.....

.....

..... [2]

(b) Describe **three** differences between solid-state and optical storage.

1 .....

.....

.....

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2 .....

.....

.....

.....

3 .....

.....

.....

.....

[6]

4 A computer stores data temporarily in random access memory (RAM).

(a) RAM is one example of primary storage.

Give **one** other example of primary storage.

..... [1]

(b) Give **one** reason why data is only stored temporarily in the RAM.

.....

..... [1]

5 A student has an image file stored on their computer.

(a) The image file was taken by a digital camera.

Tick (✓) **one** box to show whether a digital camera is an example of an input, output, process or storage device.

A input ☐

B output ☐

C process ☐

D storage ☐

[1]

6 A programmer writes a program to store and sort all the data for the books in her home library.

(a) The programmer wants to print a list of all her books on paper.

(i) Identify a suitable type of printer that she could use to print this list.

..... [1]

(ii) The programmer needs to connect the printer to her computer. The printer is on the same desk as her computer.

Give a suitable method of data transmission that could be used to connect the computer to the printer.

Explain your choice.

Data transmission method .....

Explanation .....

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.....

[6]

(iii) An even parity check is used to detect errors in the data after it has been transmitted to the printer.

Describe how an even parity check detects an error.

.....

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.....

.....

.....

..... [4]

7 An employee uses a computer that has a Von Neumann architecture to complete many different tasks in their working day.

(a) The computer has a central processing unit (CPU).

Complete the table of components in the CPU and their descriptions.

Component	Description
.....	It sends signals to all the components in the CPU to manage the flow of data through the CPU.
.....	It carries out all the arithmetic and logic operations in the CPU.
cache	....
program counter (PC)	....
.....	It controls the number of fetch–decode–execute (FDE) cycles that are performed per second.
.....	It stores data immediately before it is transmitted to RAM and immediately after it is received from RAM.

[6]

(b) The computer is an example of a general purpose computer and **not** an embedded system.

Explain why the computer is **not** an embedded system.

.....

.....

.....

..... [2]

8 A smartphone has a quad-core 2.4 GHz processor.

(a) The processor performs the fetch-decode-execute (FDE) cycle.

(i) The first register used in the FDE cycle stores the location in random access memory (RAM) of the next instruction.

Identify the name of this register.

..... [1]

(ii) Describe the actions that take place in the decode stage of the FDE cycle.

.....

.....

.....

.....

.....

..... [3]

(iii) Give the name of **one** component that is used in the execute stage of the FDE cycle.

..... [1]

(b) Explain the advantages of the smartphone having a quad-core processor instead of a dual-core processor.

.....

.....

.....

..... [2]

(c) The smartphone has built-in input and output devices.

(i) Identify **two** input devices that could be built into the smartphone.

1 .....

2 .....
[2]

(ii) Identify **two** output devices that could be built into the smartphone.

1 .....

2 .....
[2]

9 An optical mouse is connected to a computer using a universal serial bus (USB) cable.

(a) State the purpose of the optical mouse.

.....

..... [1]

10 A smartwatch has a built-in microphone that allows a person to use voice commands to control the smartwatch.

(a) Tick (✓) **one** box to show whether a microphone is an input, output, process or storage device.

A input ☐

B output ☐

C process ☐

D storage ☐

[1]

(b) Identify **one** example of a sensor that would be built into the smartwatch.

..... [1]

11 A computer has a central processing unit (CPU).

(a) State the role of the CPU.

.....

..... [1]

(b) The memory address register (MAR) and the memory data register (MDR) are two registers built into the CPU.

Give **two** other registers that are built into the CPU.

1 .....

2 .....
[2]

(c) A value is fetched from random access memory (RAM) to the CPU.

Explain how the MAR and MDR are used in this process.

.....

.....

.....

.....

.....

..... [3]

(d) The CPU has a clock speed of 3 GHz.

Describe what is meant by a 3 GHz clock speed.

.....

.....

.....

..... [2]

12 A video streaming service uses hexadecimal to display error codes when problems occur with their videos.

(a) The web server uses magnetic storage.

(i) One feature of magnetic storage is that it has moving parts.

Give **two** other features of magnetic storage.

1 .....

2 .....
[2]

(ii) The video streaming service wants faster read and write speeds for their videos than is provided by magnetic storage.

State which type of storage they should use.

..... [1]

13 A student has an application (app) on their smartphone that can be used to identify the name of a flower.

The student opens the app and takes a photograph of a flower. The photograph is stored as an image.

The app searches its database for similar images to find the name of the flower.

(a) The app uses data stored in cloud storage.

(i) Give **one** advantage to the user of the app storing data in cloud storage instead of locally on the smartphone.

.....

..... [1]

(ii) Give **two** disadvantages to the user of the app storing data in cloud storage instead of locally on the smartphone.

1 .....

.....

2 .....

.....
[2]

END OF QUESTION PAPER

Source of each question (for marking)

Find the matching official Cambridge mark scheme for the series/paper shown, to mark your answers.

Q	Past paper	Original question	Marks
1	May/June 2024 Paper 12	Q5 (a),(b)	8
2	May/June 2024 Paper 12	Q7 (a),(b)	4
3	May/June 2024 Paper 12	Q9 (a),(b)	8
4	May/June 2025 Paper 12	Q1 (a),(b)	2
5	May/June 2025 Paper 12	Q2 (a)	1
6	May/June 2025 Paper 12	Q3 (c)	11
7	May/June 2025 Paper 12	Q4 (a),(b)	8
8	May/June 2026 Paper 11	Q2 (a),(b),(c)	11
9	May/June 2026 Paper 11	Q4 (a)	1
10	May/June 2026 Paper 12	Q1 (a),(d)	2
11	May/June 2026 Paper 12	Q2 (a),(b),(c),(d)	8
12	May/June 2026 Paper 12	Q3 (f)	3
13	May/June 2026 Paper 12	Q6 (c)	3

Total: 70 marks